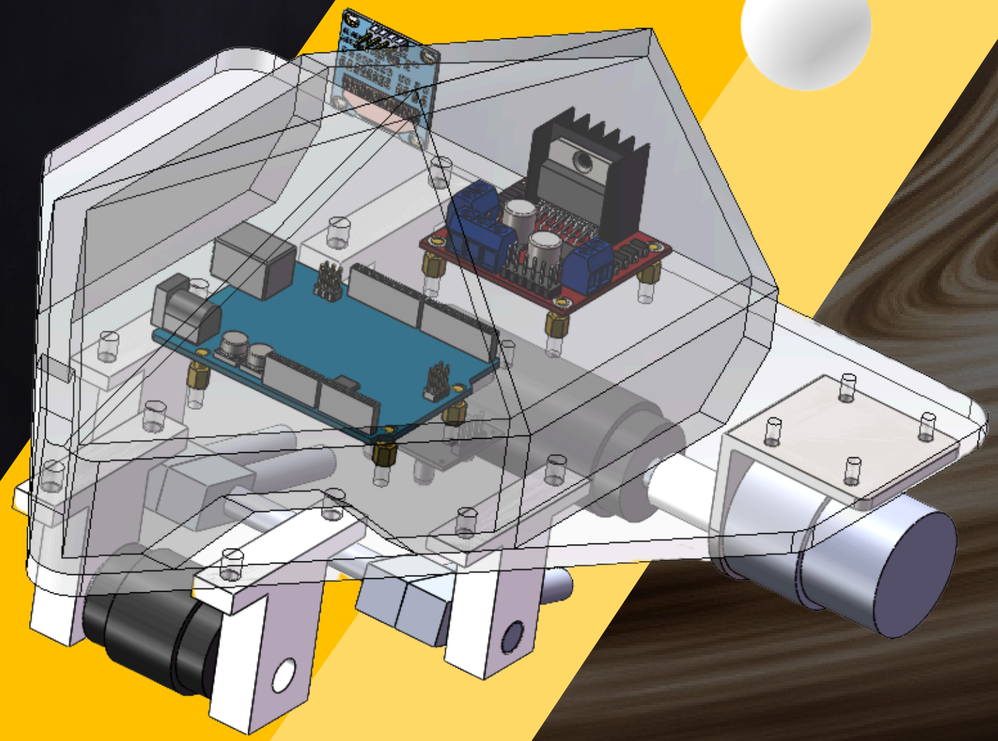


FINAL PRESENTATION

Millennium Falcon

Group 10

Wei Yushan Zhou Yifan Su Runyang
Zhou Hengyi Liu Dachuan

Team Introduction – Mutineer



周恒毅
Zhou Hengyi

Mechanical Design



刘大川
Liu Dachuan

Electric control



周亦凡
Zhou Yifan

Electric control



苏润阳
Su Runyang

Mechanical Design



卫宇珊
Wei Yushan

Mechanical Design

Contents

01 Design of the Final Concept

02 Selection of Materials and Components

03 Prototype Assembling

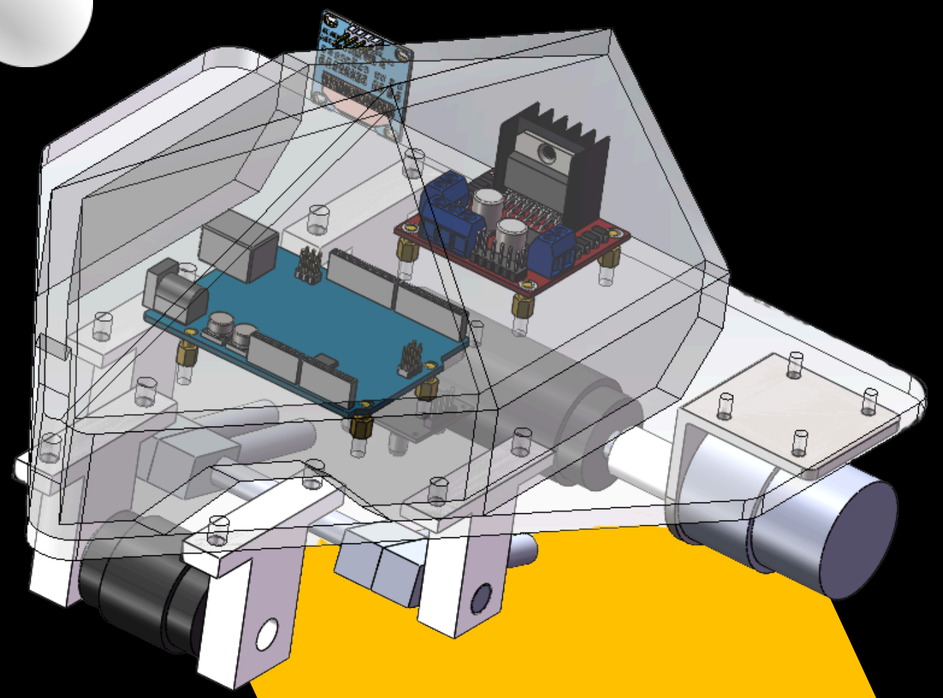
04 Design of the Control System

05 Video, Cost & Discussion

Group 10

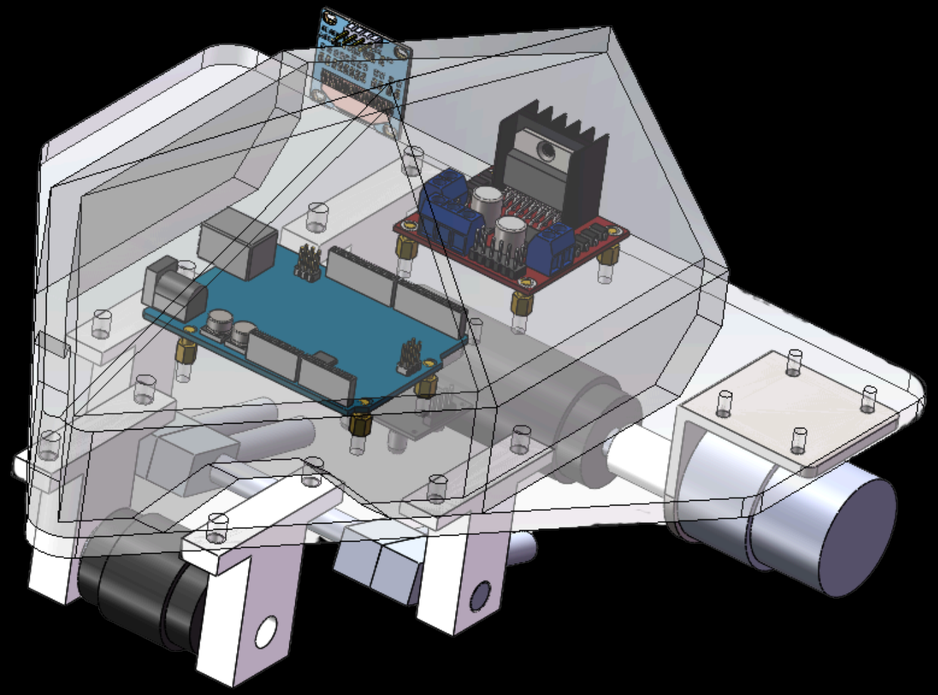
PART ONE

Design of the Final Concept



Final Design

A back wheel driven, gravity based suspended rod vehicle ---- The Millennium Falcon.



Driving System

Passive Limiting System

Controlling System

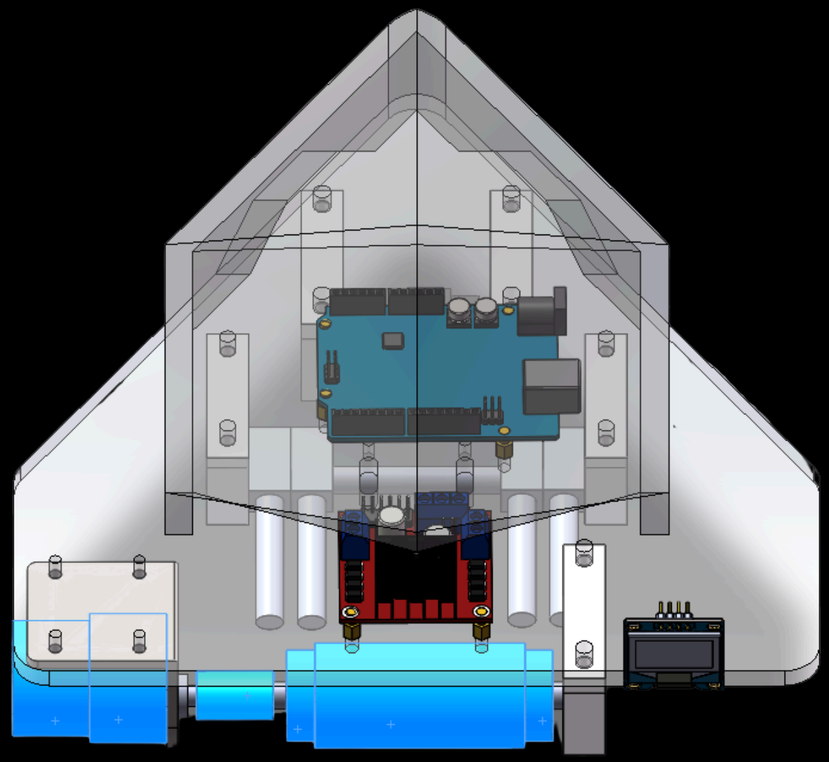
Driving System

55 RPM Motor:

- large torque
- direct connect for simplicity
- (counterweights on the right)

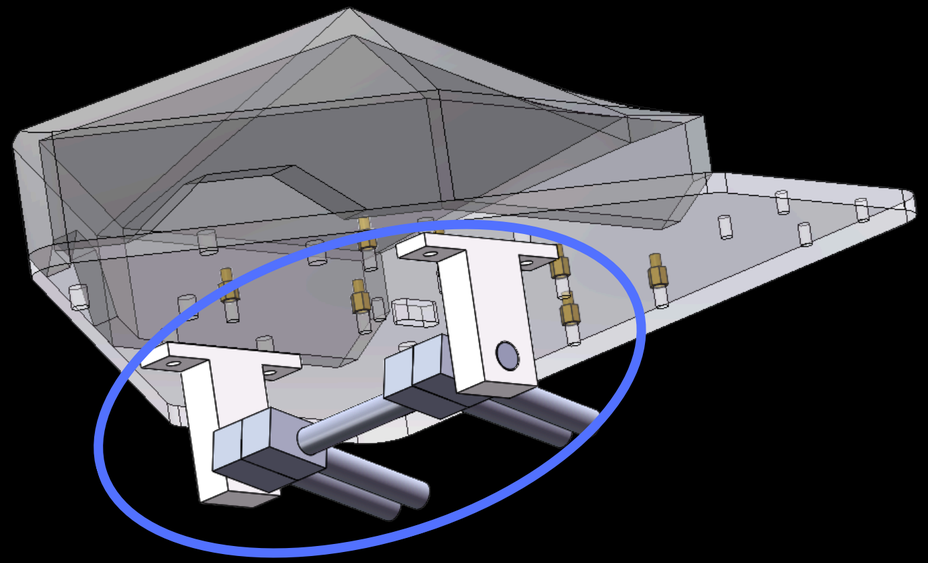
Polyurethane Roller:

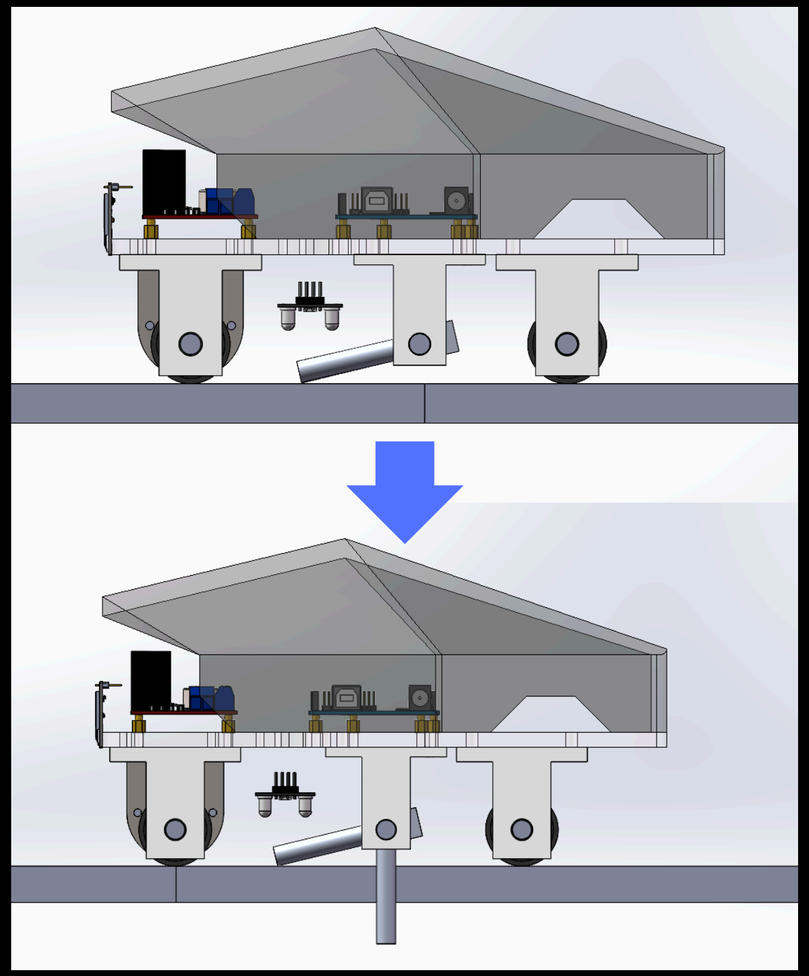
- prevents slippery



Passive Limiting System

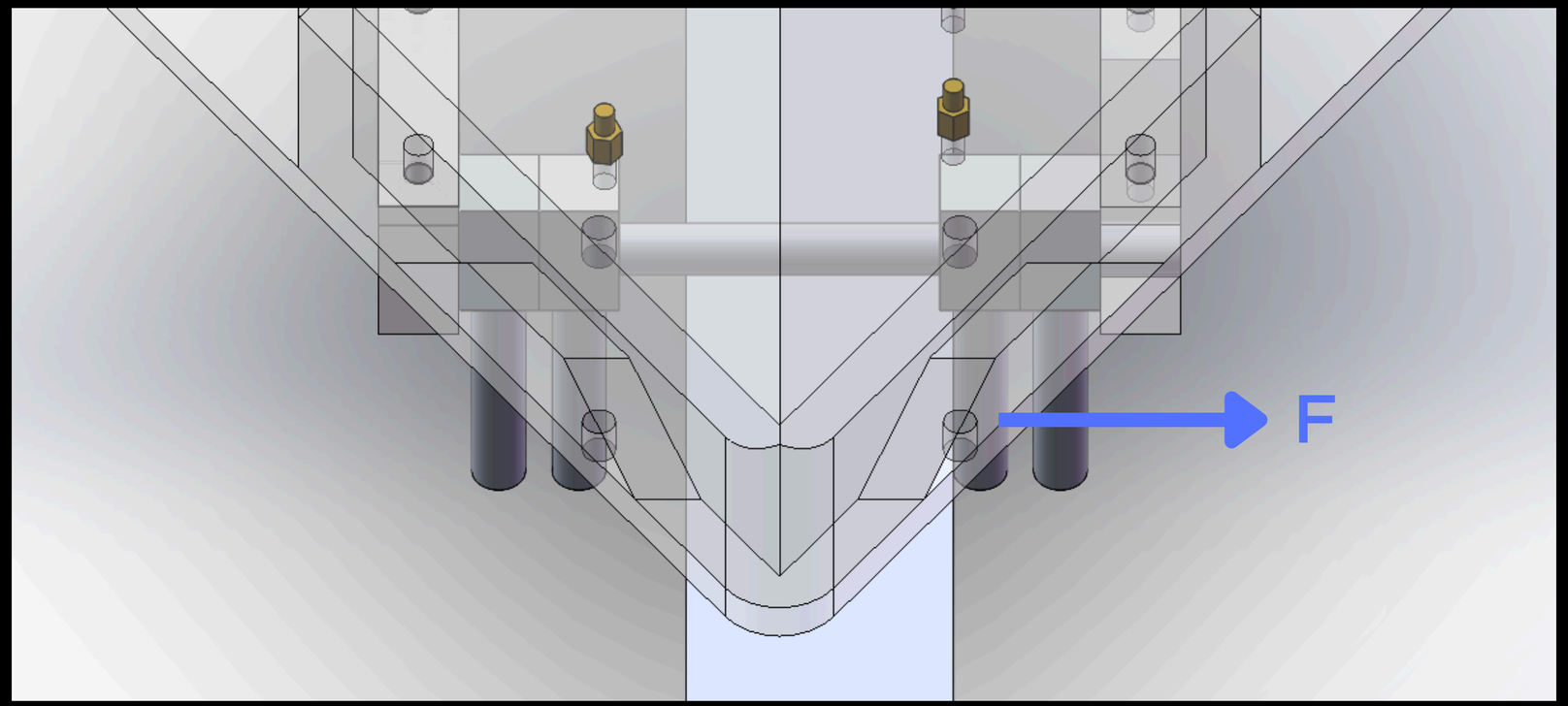
- 4 suspended rods
- 40 mm spacing*2; 60 mm spacing*2
- Free rotate, no power supply
- Hang down when passing the bridge





Passive Limiting System

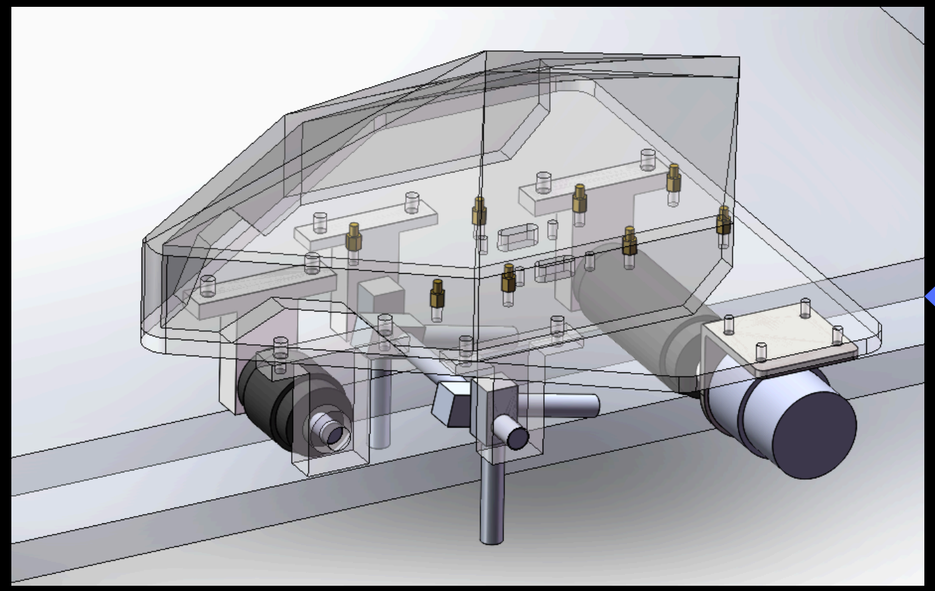
Supporting force by the edge of the bridge



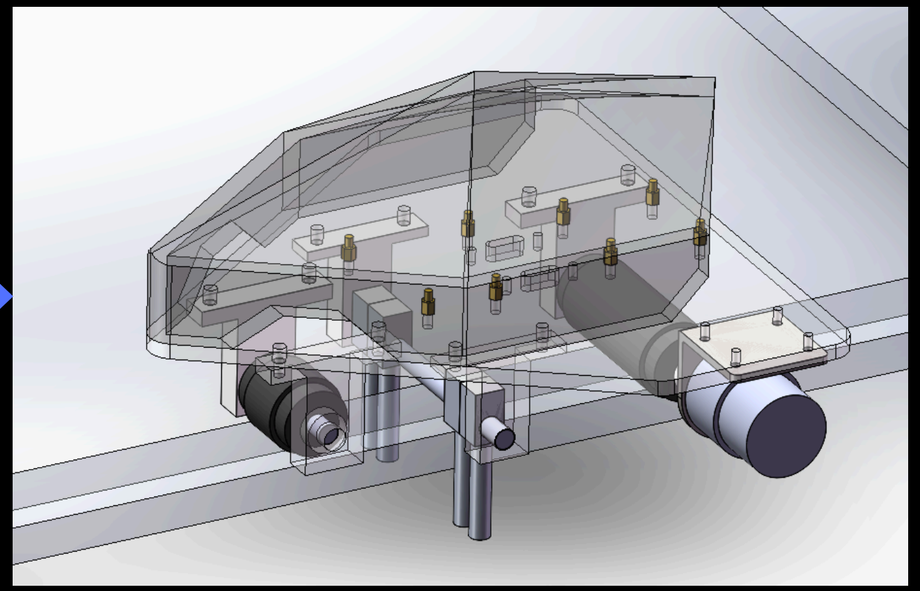
Passive Limiting System

Adaptability among bridges:

60mm, Only outer rods down

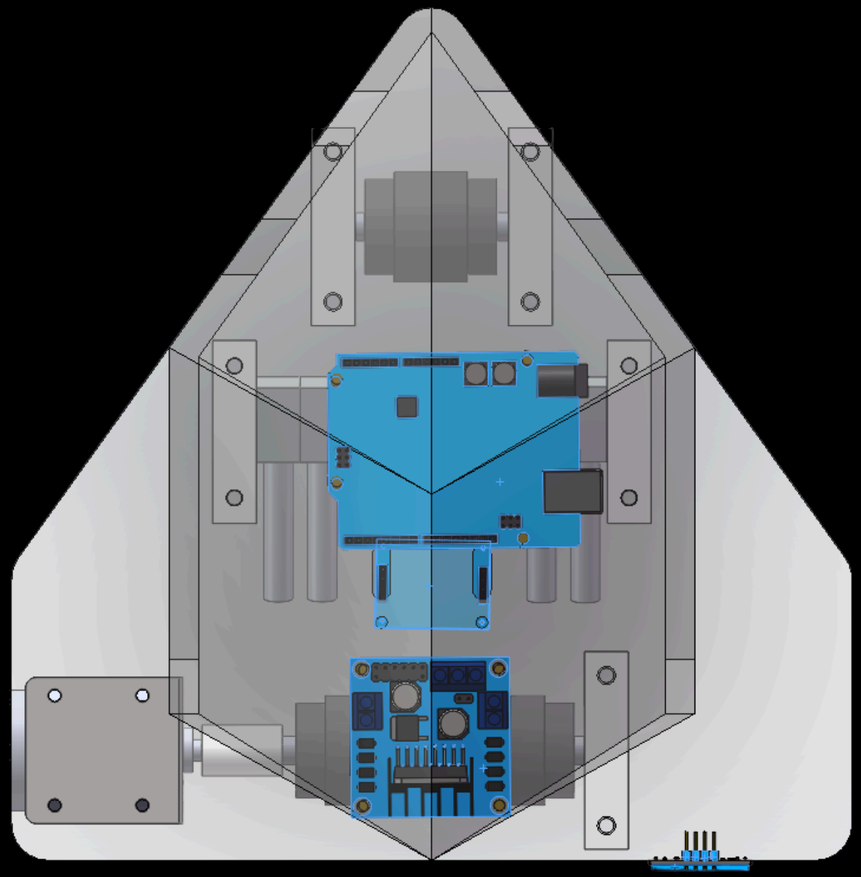


40mm, Both rods down



Controlling System

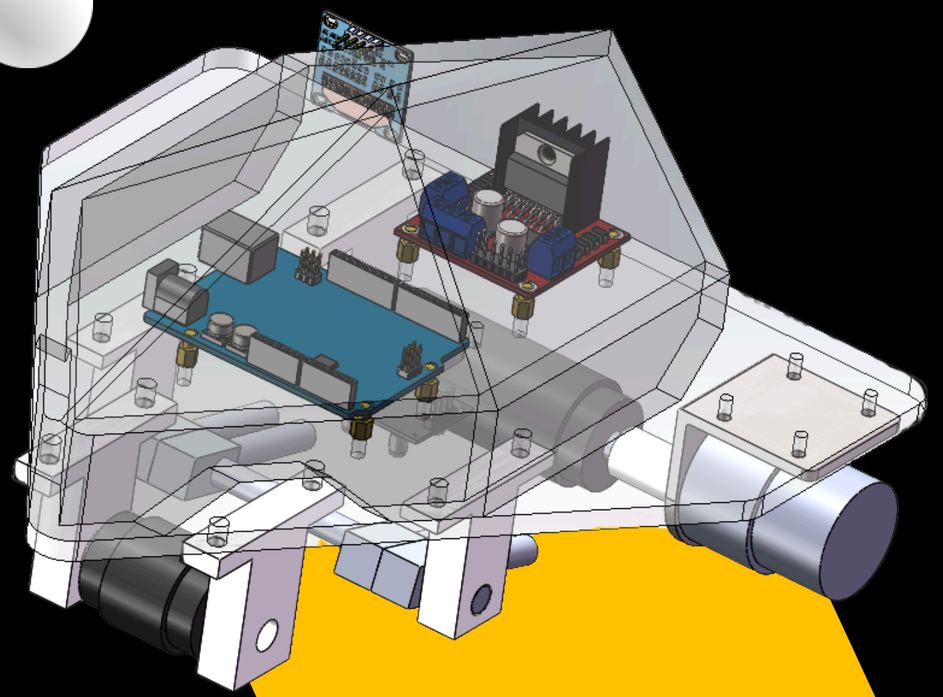
- Arduino board
- L298N board
- Color sensor
- LED display screen



Group 10

PART TWO

Selection of Materials and Components



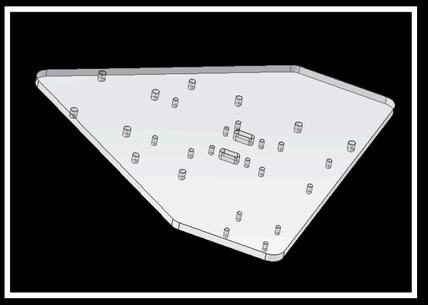
Material Properties

Lightweight

Strength

Durability

Parts	Material
Base Plate	**Acrylic**
Supporting Column	Copper
Outer Shell	Polylactic Acid (PLA)
T-shaped Bracket	Polylactic Acid (PLA)
Wheel Axle	Stainless Steel
Sleeve Coupling	**Stainless Steel**
Suspended Rod	Aluminum Alloy
Axle Sleeve	**Polyurethane (PU)**

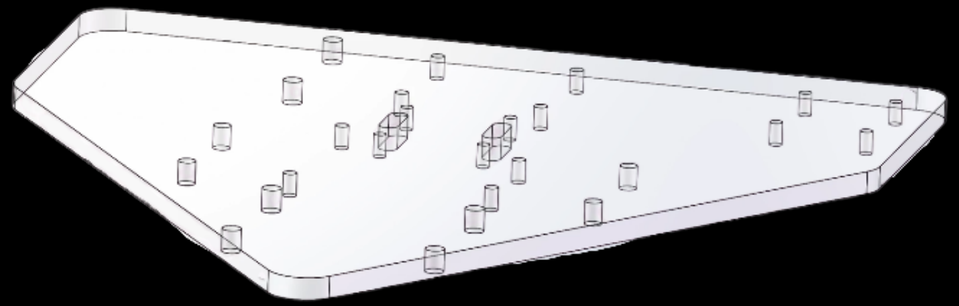


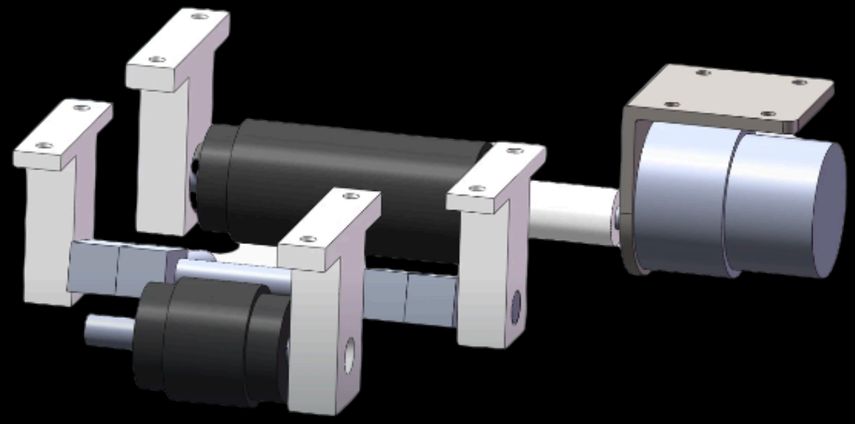




Material Properties

Parts	Material
Base Plate	Acrylic
Supporting Column	Copper
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T-shaped Bracket	Polylactic Acid (PLA)
Wheel Axle	Stainless Steel
Sleeve Coupling	Stainless Steel
Suspended Rod	Aluminum Alloy
Axle Sleeve	Polyurethane (PU)





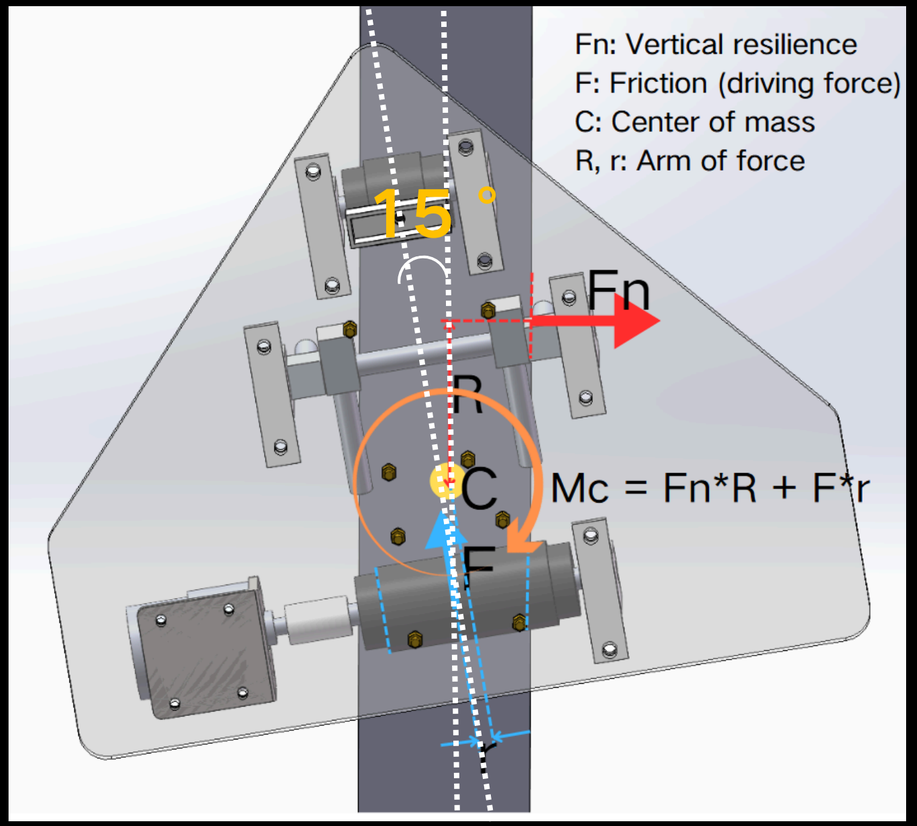
Statics Analysis of Post-offset Correction

If the vehicle turns left,

There would be a reaction force *Fn* exerted by the bridge on the (right) rod.

$$M_C = F_n \cdot R + F \cdot r$$

A reverse torque is created to correct body posture.



Finite Element Analysis

Given:

- Total mass of the car, $m = 1600\,\text{g} = 1.6\,\text{kg}$
- Distance from front wheel to center of mass, $d_f = 95\,\text{mm}$
- Distance from rear wheel to center of mass, $d_r = 47.5\,\text{mm}$
- Coefficient of friction for rear wheels, $\mu = 0.4$
- Radius of rear wheel, $r = 1.5\,\text{cm} = 0.015\,\text{m}$

Calculation:

The support force ratio is derived from the distances to the center of mass:

$$\frac{N_f}{N_r} = \frac{d_r}{d_f} = \frac{47.5\,\text{mm}}{95\,\text{mm}} = \frac{1}{2}$$

The total weight of the car is:

$$W = m \cdot g = 1.6\,\text{kg} \times 9.81\,\text{m/s}^2 = 15.696\,\text{N}$$

The system of equations to solve for the rear wheel torque is:

$$\begin{cases} N_f + N_r = W \\ \frac{N_f}{N_r} = \frac{1}{2} \end{cases}$$

Solving for N_r:

$$N_r = \frac{2}{3} \cdot W = \frac{2}{3} \cdot 15.696\,\text{N} = 10.464\,\text{N}$$

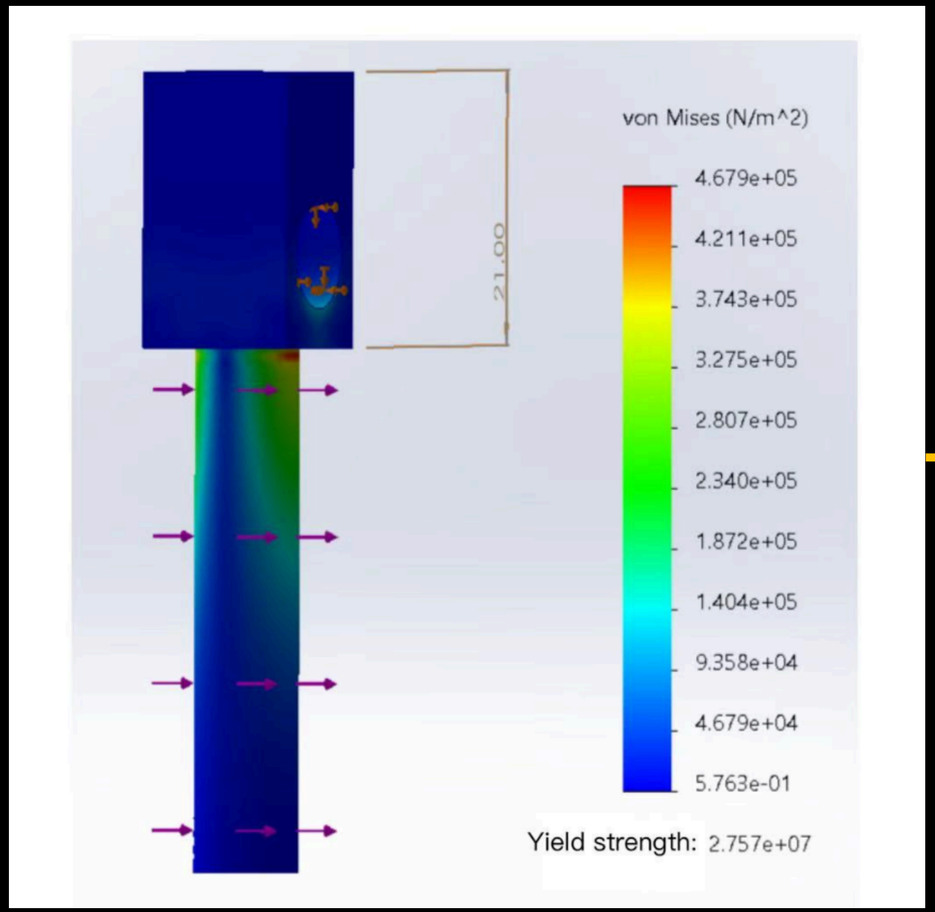
The friction force on the rear wheels is:

$$F = \mu \cdot N_r = 0.4 \cdot 10.464\,\text{N} = 4.1856\,\text{N}$$

Here we suppose the deviation angle to be 15°,

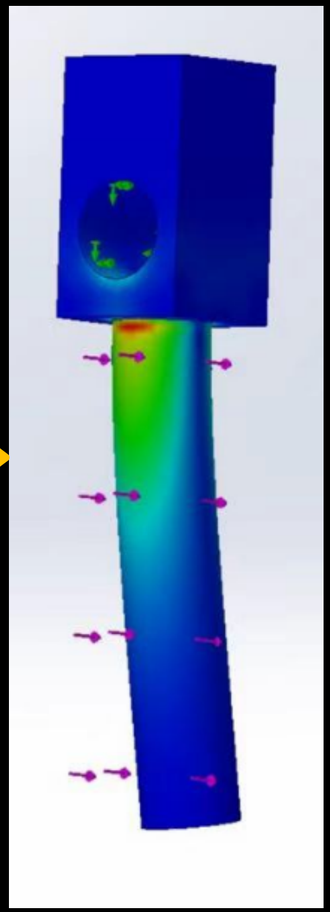
$$F_{\text{rod}} = F \sin(15°) = 1.08N$$

Finite Element Analysis



1060 Aluminum Alloy
F = 1.08 N

Scale value = 8920



Motor Analysis

2.5 Calculation of Rotational Speed n

The linear velocity v of the wheel is calculated as:

$$v = \frac{\text{distance}}{\text{time}} = \frac{0.7\,\text{m}}{9.945\,\text{s}} \approx 0.0704\,\text{m}\,\text{s}^{-1} m/s$$

The angular velocity ω is given by:

$$\omega = \frac{v}{r} = \frac{0.0704\,\text{m/s}}{0.015\,\text{m}} \approx 4.693\,\text{rad}\,\text{s}^{-1} rad/s$$

The rotational speed n in rpm is:

$$n = \frac{\omega \cdot 60}{2\pi} = \frac{4.693 \cdot 60}{2 \cdot 3.14159} \approx 44.83 rev/min$$

2.6 Operation Line

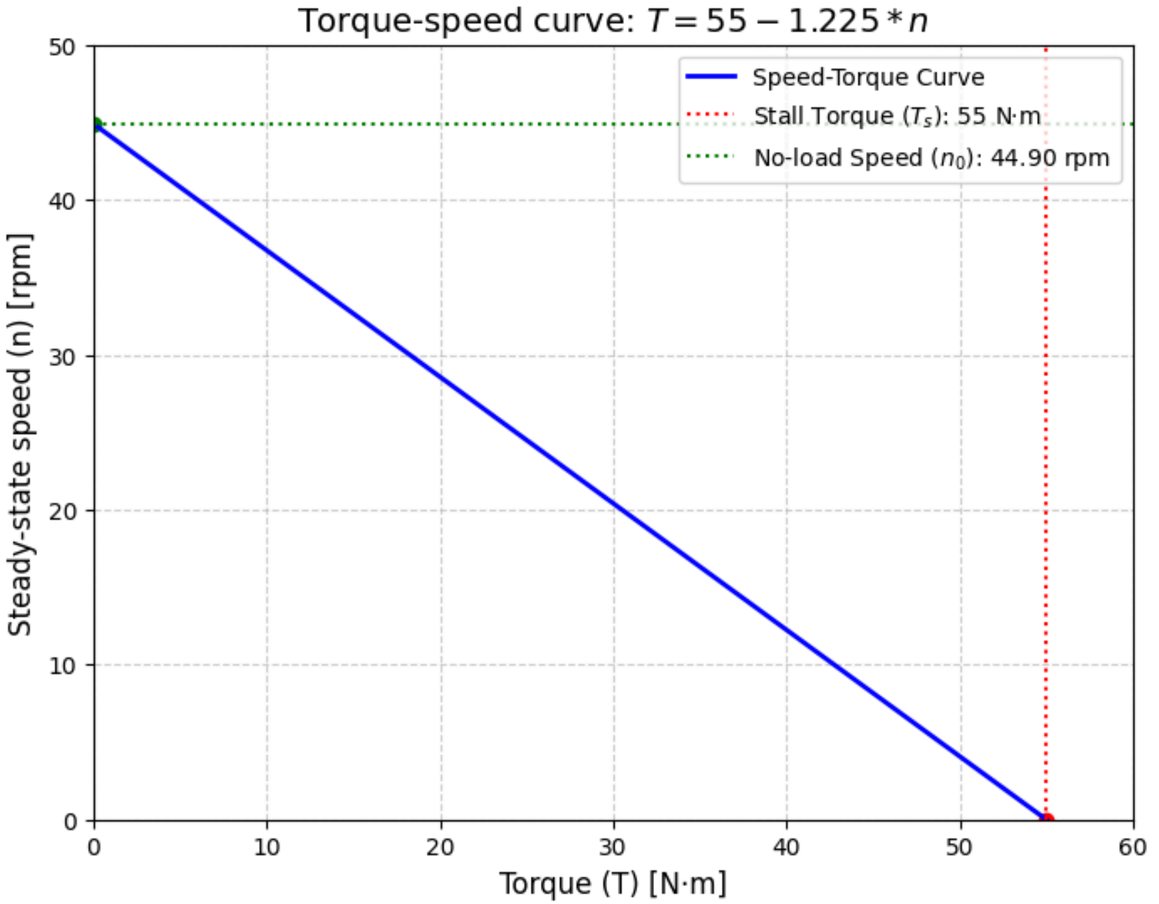
In our rotary motor system, the parameters γ and M are both equal to 1, i.e., $\gamma = 1$ and $M = 1$. Thus, $T'_{sg} = \gamma T_{sg} = \gamma M T_s$ simplifies to:

$$T'_{sg} = T_{sg} = T_s$$

Therefore, the line given by the formula $T = T_s - kn$ passes through points $(0, 55\,\text{N m})$ and $(0.0628, 44.83)$.

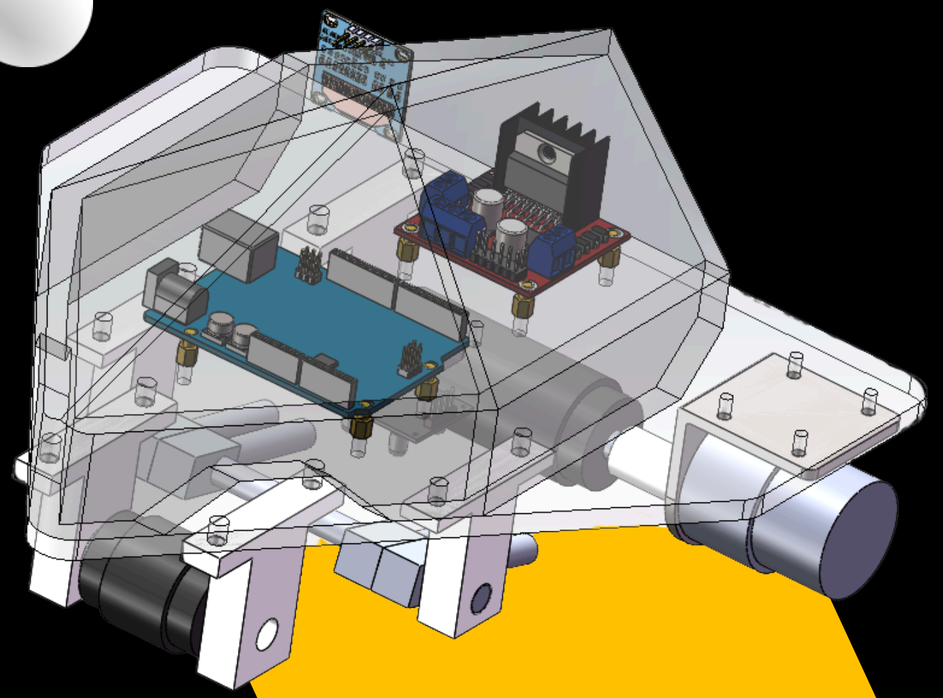
The operation line equation is:

$$T = 55 - 1.225n$$

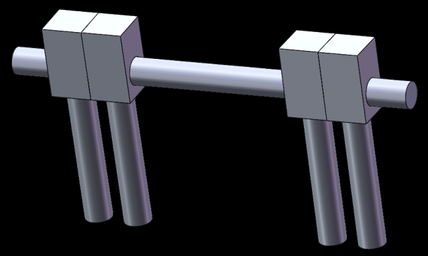


PART THREE

Prototype Assembling



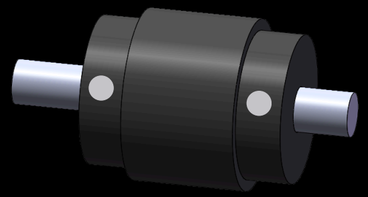
Assembly Flow

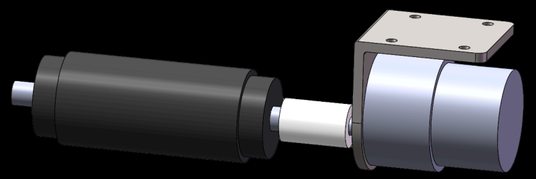


Using:

- suspended rods
- rubber band (between rods)

Axle connection → Screw and bearing fixation → Circuit components installation → Monitor and casing installation





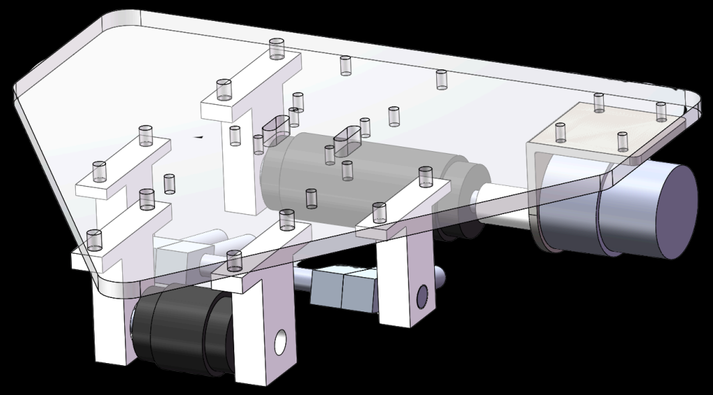
- wheels
- thread
- coupling

Assembly Flow

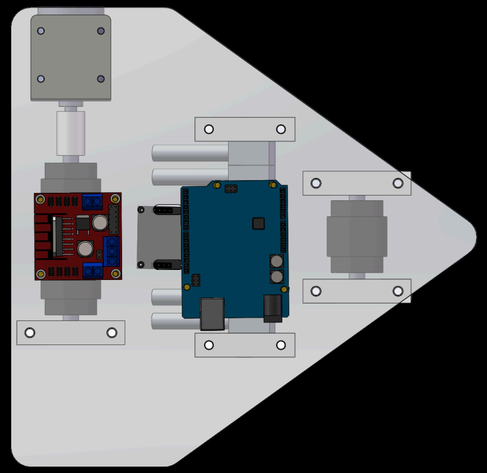
Using:

- T-shaped adapter
- motor holder
- M4 screws

Axle connection → **Screw and bearing fixation** → Circuit components installation → Monitor and casing installation



Assembly Flow



Axle connection → Screw and bearing fixation → Circuit components installation → Monitor and casing installation

Using:

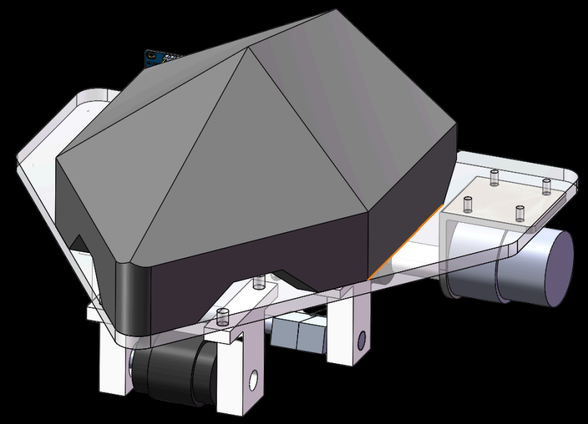
- Wires
- Copper columns
- Velco strap (for easy installation and removal)



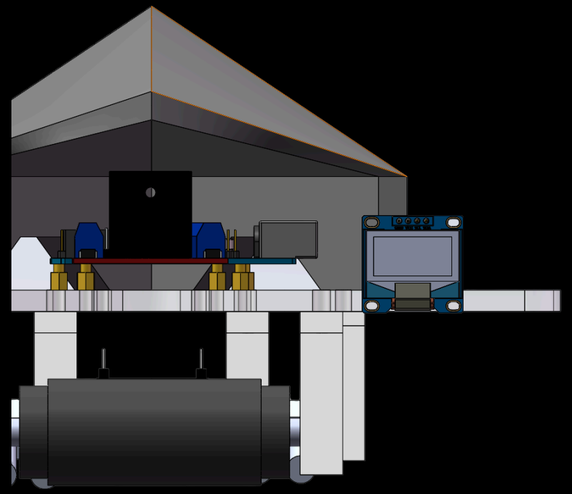
Assembly Flow

Using:

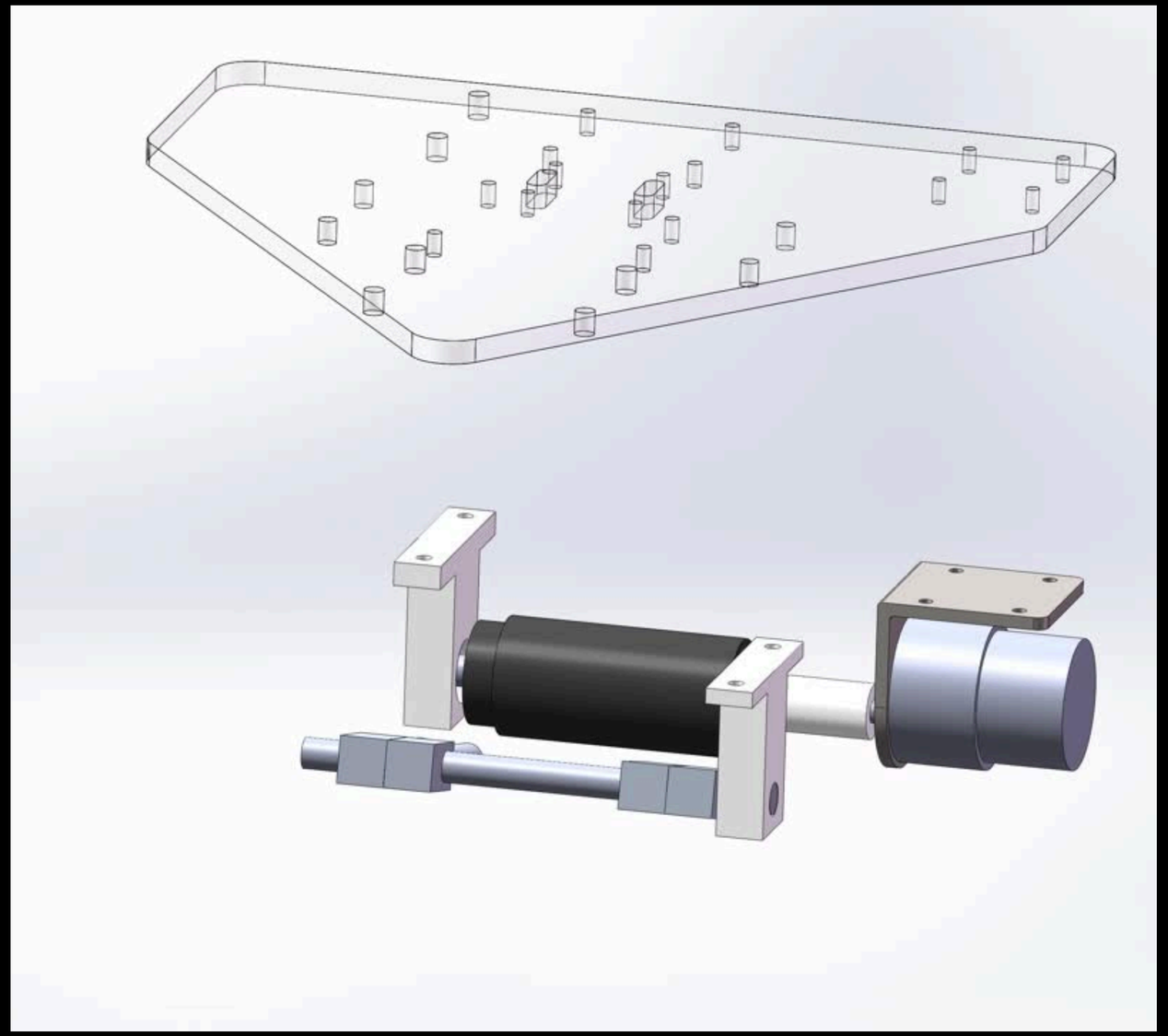
- Velco strap



Axle connection → Screw and bearing fixation → Circuit components installation → Monitor and casing installation

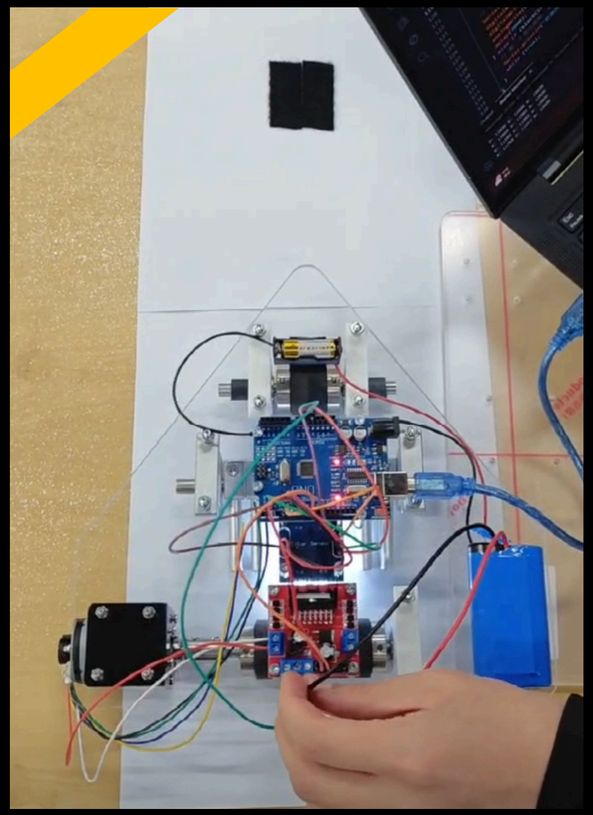


Exploded View



Manufacturing Tests

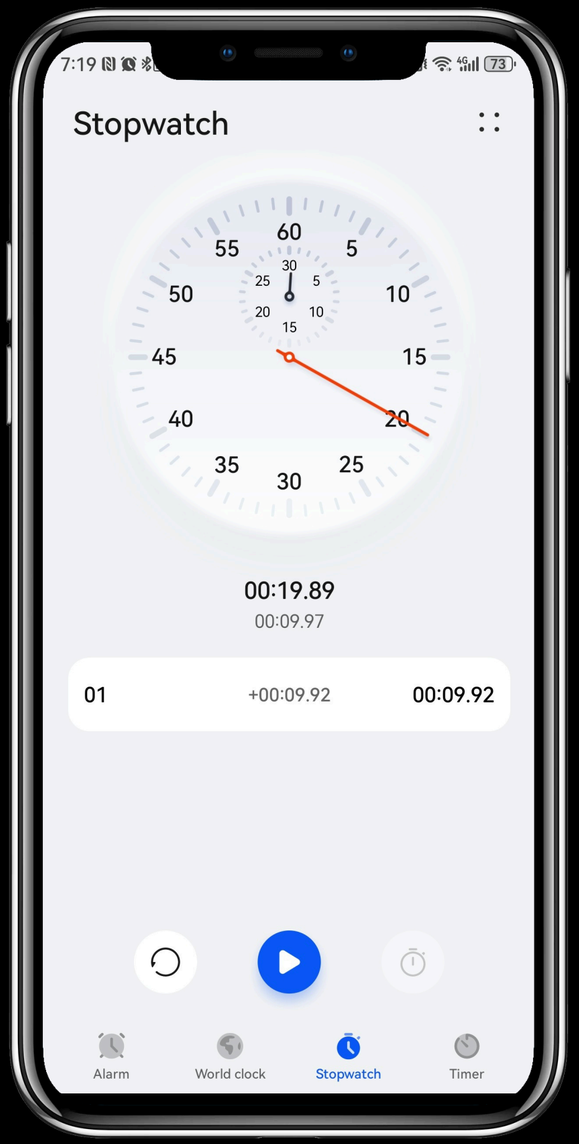
- Straightness of the trajectory



▲ The vehicle deviates toward the right because of larger friction on the left.



▲ Another weight added to the battery side.



Manufacturing Tests

- Passing Time

60mm

9.92s

From the yellow mark to the next yellow mark.

40mm

9.97s

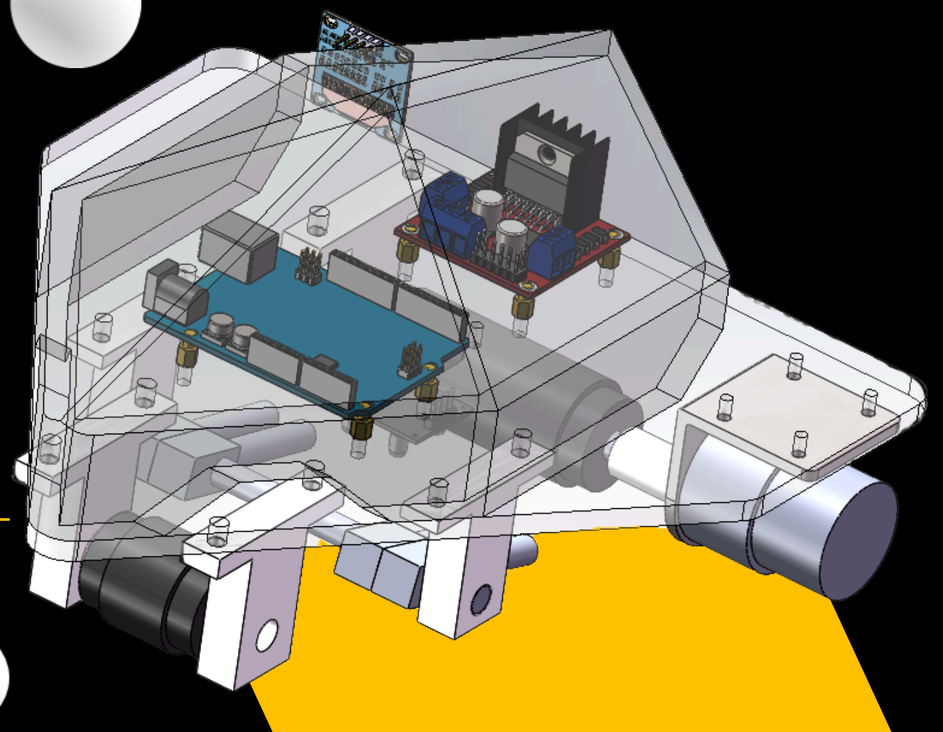
From the yellow mark to the next yellow mark.

Requirement: 120s

Our Total Time: 19.89s

PART FOUR

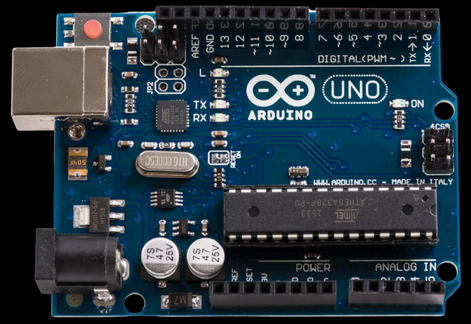
Design of the Control System



Control Boards

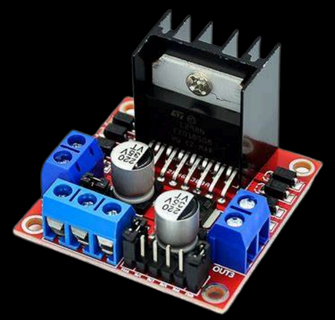
Arduino Uno

Coordinate All the Components



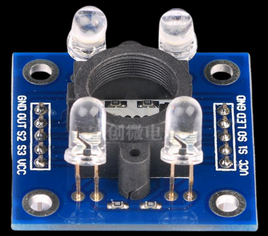
L298N

Motor Power Delivery



Color Sensor

Detect Yellow Color : R > 7000

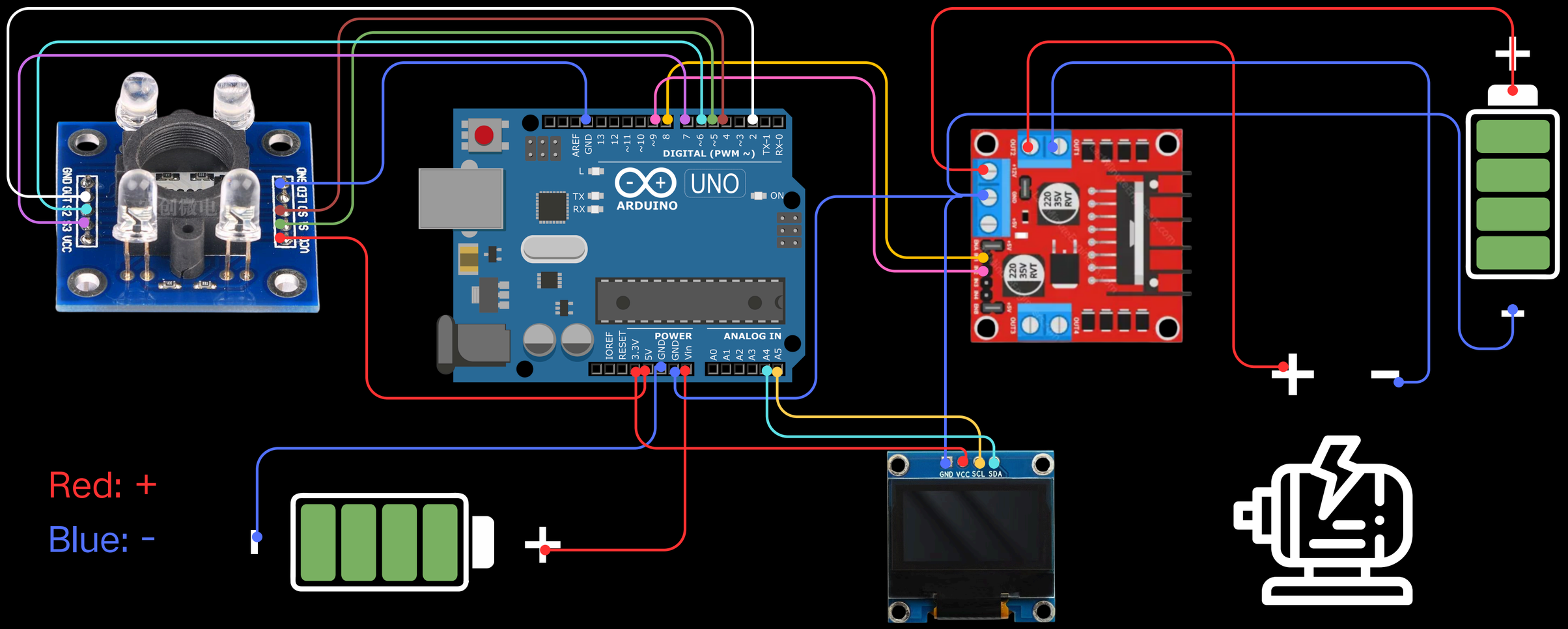


LED

Fast Debug without Connecting with PC

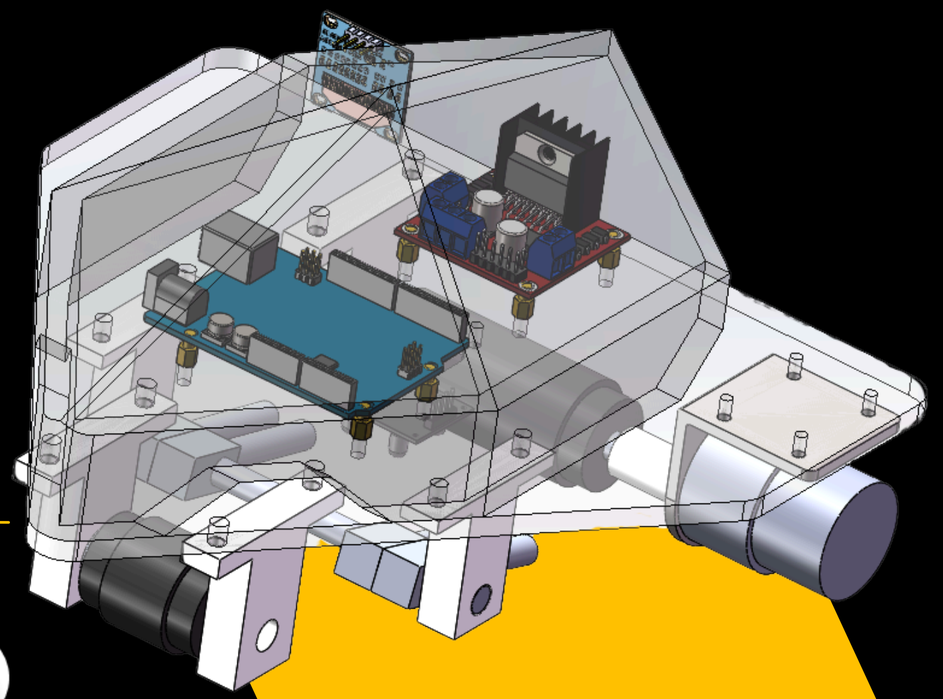


Circuit Diagram

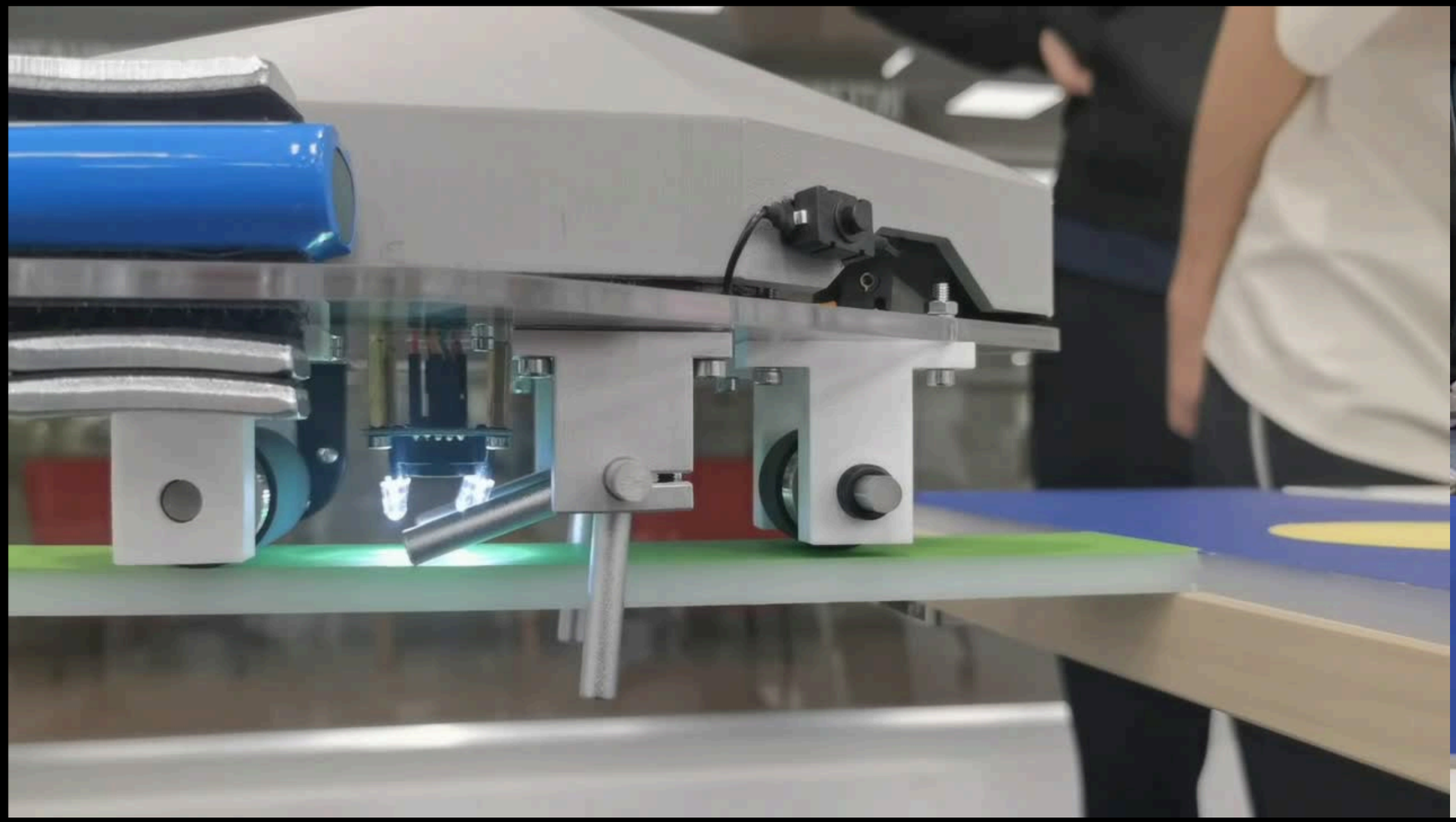


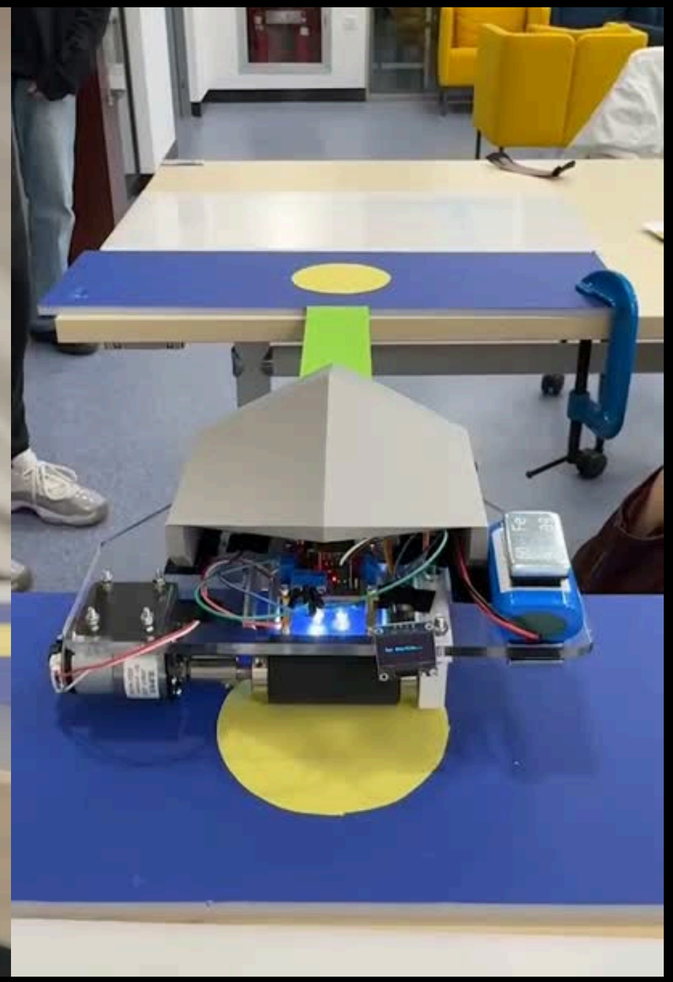
PART FIVE

Video, Cost & Discussion

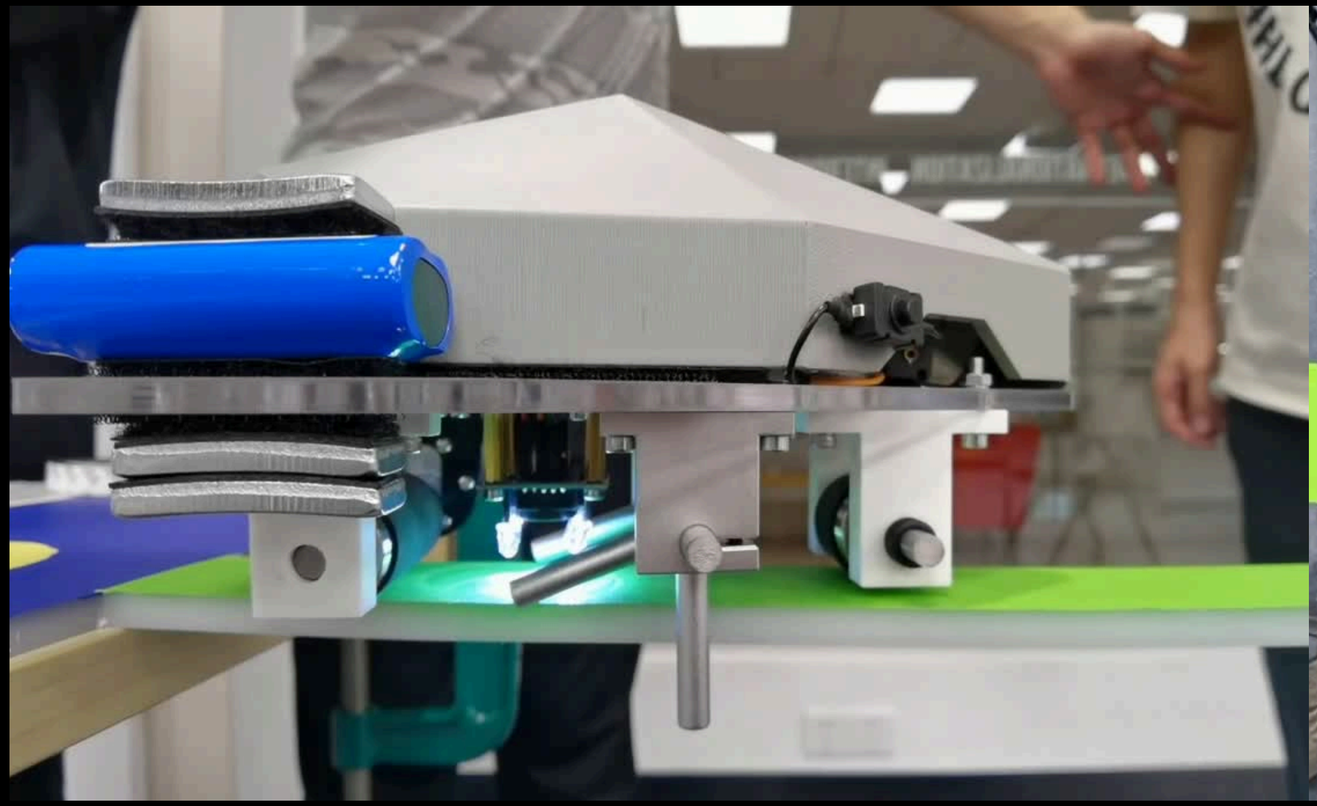


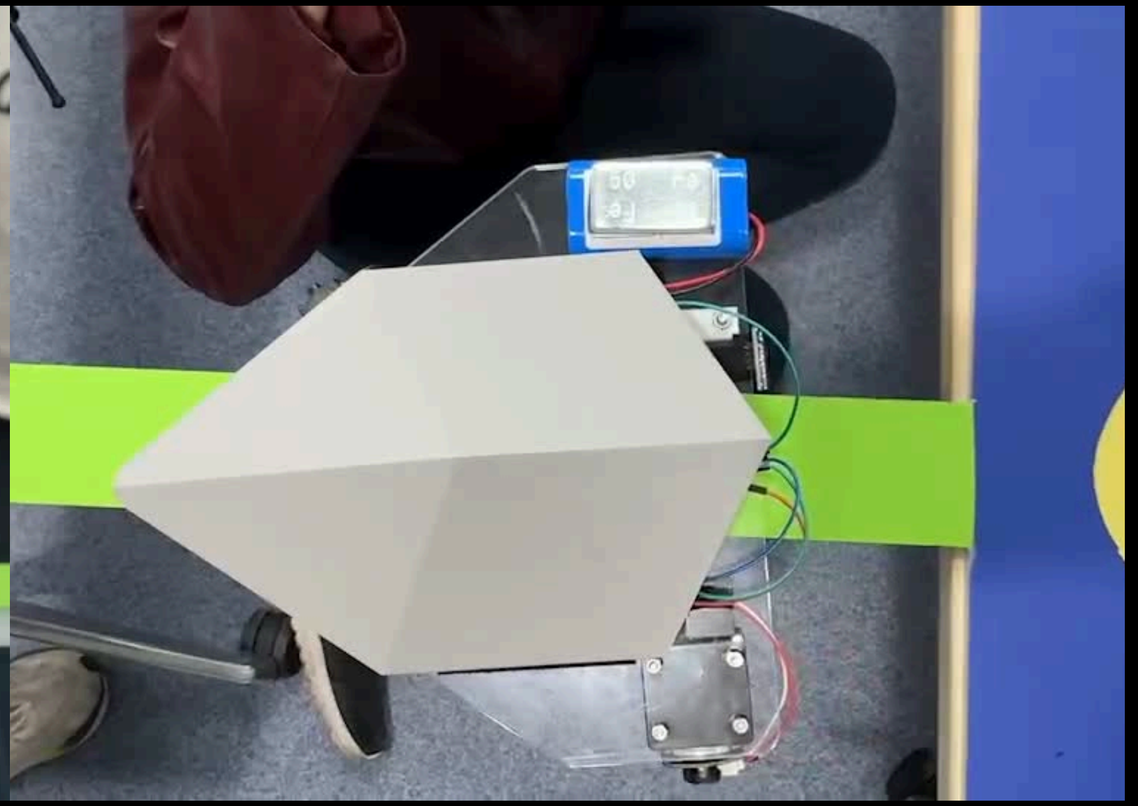
Video (complete process)





Video (when deviation occurs)





BOM					
Item	Model / Spec	Quantity	Unit Price (yuan)	Total Price (yuan)	Link
Arduino	UNO	1	0	0	/
Motor Driver	LN289	1	0	0	/
Color Sensor	TCS3200	1	18.8	18.8	LINK
LED Display	0.96'	1	13.8	13.8	LINK
Infrared Sensor	TCRT5000	1	4	4	LINK
Acrylic Plate	6mm	1	0	0	SIC
Copper Pillar	M3 * 5 + 4 M5 * 8 + 7	20 10	0	0	SIC
Casing	/	1	0	0	SIC
T Bracket	/	1	0	0	SIC
Front and Rear Wheels	Rear: 30mm * 60mm - 8mm Front: 30mm * 20mm -8mm	2	38 + 29	67	LINK
Axle	8mm * 100mm8mm * 150mm	2	1	2	LINK
Motor	55RPM 6V	1	37	37	LINK
Coupler	6 to 8	2	4	8	LINK
T AdapterHanging Shaft	SHWTA8-30	2	20	40	LINK
Universal Joint Rod (Hanging)	8mm * 40mm length	4	7.5	30	LINK
Battery Holder	12V single-cell with wires	1	2.5	2.5	LINK
Battery	Buy several	1	2	2	LINK
Rubber Sleeve	Inner diameter 8mm outer 12mm	1m	4	4	LINK
Velcro	2cm * 3m	1	15	15	LINK
Counterweight	50g	3	1	3	LINK
			TOTAL:	247.1	

Discussion

Advantage

Adaptive Passive Limiting System

Durable Material

Fun & Attractive Appearance

Effective Control System

Disadvantage

Passive Limiting Constraint

The rods only has limiting effect when the width of the bridge is 40 and 60mm.

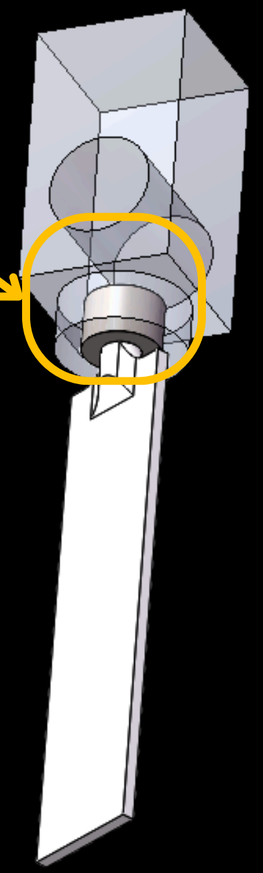
Inadaptive Limiting Rods

The rods can not restrict car’s position with large deviation.

Further Development

More Adaptive Suspended Rod

Our current design is hard to save the car from falling if the deviation angle is too large. With one more bearing, it will be more adaptive to deal with larger deviation angle.

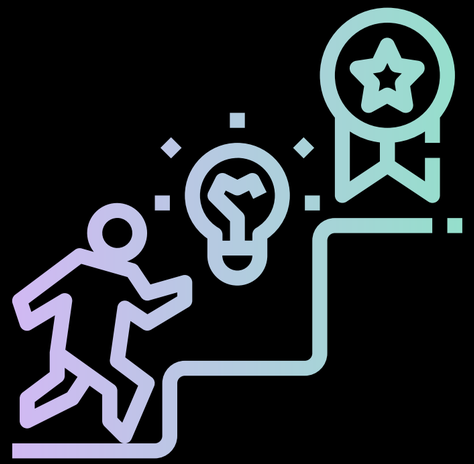


The Width is Continuously Adjustable

Our current design can only deal with the bridges of 60mm and 40mm. With the width of the suspend rod continuously adjustable, the car can cope with bridges of different width easily.

Recommendation

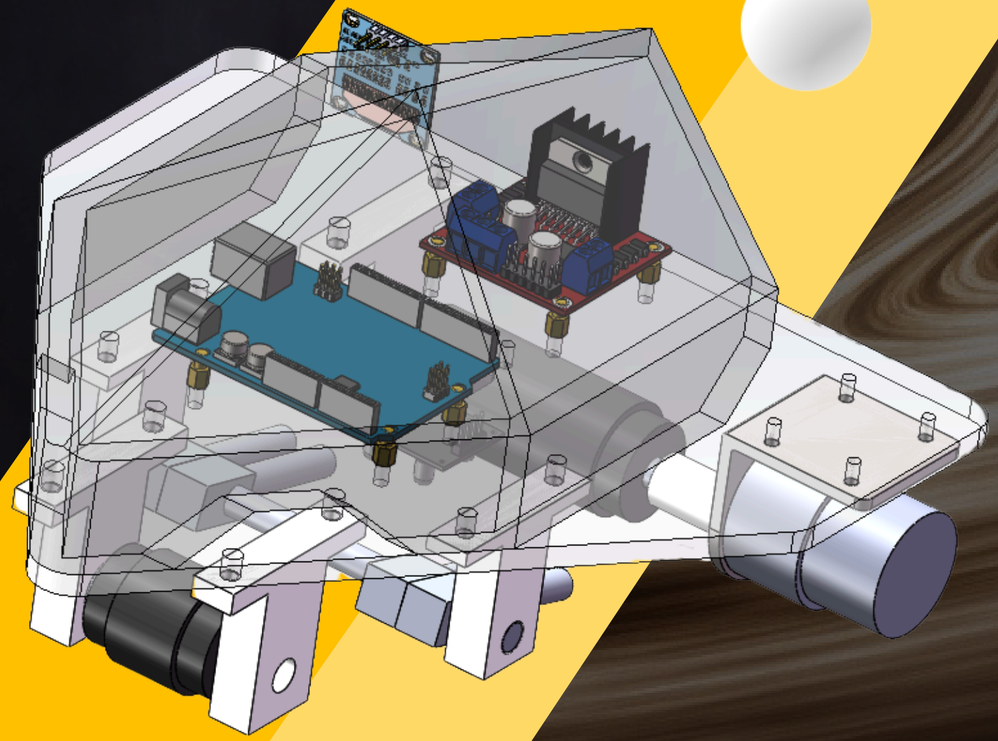
- Think further instead of just few upcoming steps.
- Be more efficient.
- Be attemptive. Try more.



Reference

[1] ME2500J Project Description

[2] https://www.cgmol.com/model/6/96109.html



THANK YOU!

Q&A

Millennium Falcon

Group 10

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